Sahaj Bhadja

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Summary

Software engineer with 2+ years building **backend systems**, **REST APIs**, relational data pipelines, and **multithreaded** applications using Python, Java, SQL, cloud platforms, and rigorous automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Database Management Systems, Cloud Computing, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built recurring data integrations using FastAPI, SQL Server, and cloud storage, processing 5K+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases at 99%+ reliability in production environments.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed a **concurrent batch-processing** pipeline using multithreading and secure PDF, DOCX, and ZIP ingestion through AWS RDS and REST APIs, processing 1K+ files per batch reliably under concurrent workloads.
- Implemented an AWS-hosted backend serving 150+ instructors and 10K+ daily requests, using CI/CD-driven workflows and model-usage tracking to reduce feedback turnaround time by 60% while maintaining peak-load stability.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a quantitative Decision Transformer and end-to-end evaluation pipeline processing 100K+ trajectory logs, improving long-horizon planning by 18% through structured modeling, testing, and simulation analysis at research scale.
- Architected a spatial-temporal encoder integrating map geometry and agent histories through multi-head attention, improving closed-loop stability by 15% while evaluating complex model behavior and observing fewer off-road trajectories across simulations.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform microservices with automated builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS release environments with repeatable nightly reporting.
- Optimized large-scale **API integrations** using OOP patterns, AWS Glue, S3, and SQL, improving response time by 30%, pipeline efficiency by 68%, and storage costs by 15% for low-bandwidth users.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated payment infrastructure supporting cards, wallets, and multi-currency transactions with real-time analytics, webhook-driven refunds and disputes, and approximately 70% less manual payment handling across production transaction workflows.
- Developed Flask APIs and Firebase workflows for 50K+ booking records, delivering sub-200-millisecond responses, real-time transaction-state synchronization, and resilient pricing and coupon validation across platforms with consistent data integrity.

Skills

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, Bash
Backend & APIs: FastAPI, Flask, REST APIs, HTTP/S, authentication, microservices, multithreading, OOP, design patterns
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, SQLite, MongoDB, Firebase, relational schemas, query optimization, reconciliation
Cloud & DevOps: Azure, AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: SDLC, unit testing, integration testing, debugging, monitoring, performance tuning, Agile/Scrum, technical documentation, code review

Projects

TradeGuard | Python, SQLite, multithreading, order routing, trade reconciliation

- Built and tested a multithreaded trading service with idempotent order routing, relational persistence, notional exposure calculations, and execution reconciliation that flags quantity breaks and unknown orders during automated test scenarios.

FinNexus | Live | Next.js, TypeScript, Kubernetes, MongoDB, Plaid

- Built a scalable fintech platform supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based security, real-time synchronization, advanced filtering, and Kubernetes-backed CI/CD deployment across secure account workflows.

Chat2SQL | Code | Java, TypeScript, SQL, Docker

- Built an AI-assisted SQL workspace supporting 10+ databases, combining schema-aware query generation, structured validation, and Docker-based deployment to accelerate relational data exploration and database management from one unified interface.